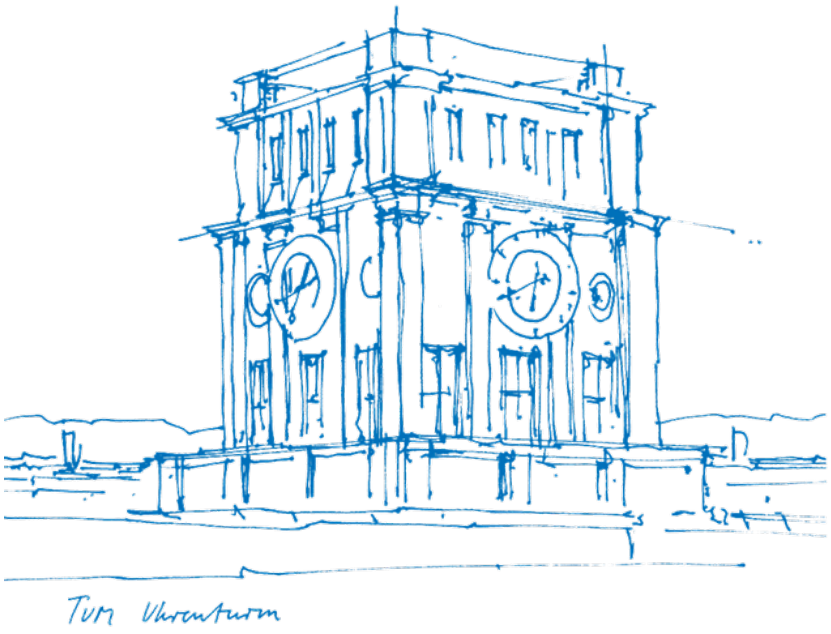
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Barycenter Approximating of Causal Effects

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Baryzentrische Approximation kausaler Effekte

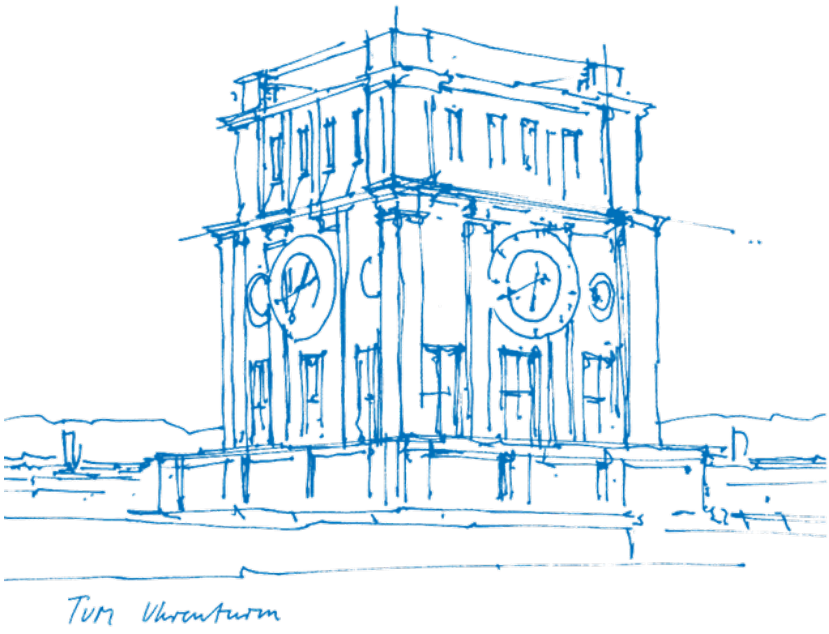
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Abstract

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Any process that accompanies human activity in various fields, from taking medicine to advertisement campaigns and crop yields, answers the question: "What happens to the outcome if an influencing factor changes?" Causal inference is the scientific field that deals with such questions; in particular, it measures the causal effect if it is identifiable in a relationship of interest.

One of the challenges of the aforementioned procedure is dealing with uncertainty of relationships between events of investigated processes in real-world scenarios. This work aims to combine the ID-algorithm with probabilistic causal graphs by approximating the causal effect with the help of Wasserstein barycenters. We show that decomposing a graph into its possible subgraphs and aggregating individual estimations of each in a form of a barycenter appears to be a good estimate of a target causal query.

Such an approach captures all possible combinations of relationships between events in a given process.

Particular observations were discovered as a result of a number of conducted experiments with involved parameters.

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Introduction

The ever-present importance of such domains as healthcare and finance, along with the expanding potential of machine learning and its integration into various decision processes, makes it essential to have a clear understanding of how each such process is built. Causality underpins any conclusion that aims to explain relationships, make predictions, assessments, and form expectations. Central to this is an understanding of the true causes of particular events in processes of interest. Such causes might not be clean: ambiguous, consist of many, or be influenced by unobserved factors.

The two main challenges of causal inference are:

1) to find the difference between two conditions of a single unit - same object being in a treatment and in a control group, making an individual causal effect never directly measurable;

2) finding the pure influence of the treatment variable on the outcome variable, based on the observational data only, where the effect can be influenced by unobserved factors.

This work focuses on the second problem: identification and estimation of causal effects in probabilistic semi-Markovian models.

Finding the relationship between events of interest, in particular, proving the causal nature of a relationship, can be realized by performing experiments and controlled interventions. Such methods of gathering the data, despite their high reliability, are hard or even, due to a number of various reasons such as practical, ethical or economic [9], impossible to organize in a multitude of cases. Observational data, on the contrary, is usually much more accessible. However, here the problem of unmeasured variables arises, which draws attention to the question of the identifiability of causal effect. This

challenge was addressed by Judea Pearl. In his research, he has concluded that the causal effect is uniquely identifiable if, using the three rules of *do*-calculus, the initial interventional probability distribution can be transformed into a *do*-free expression, containing only available observational probability distributions. [1]

The graphical representation allows the simplification of the ID task by introducing the possibility of application of the so-called front-door and back-door criteria [1]. A respective causal graph illustrates the relationships between variables in the model. The causal discovery problem deals with learning such graphs from the available data. Since causal discovery is data-dependent and noisy, the structure of a graph is uncertain. Thus, we are dealing with a probabilistic graph, which means that the causal effect might be identifiable for some versions of such a graph and not for others, as well as require different estimation strategies.

In order to deal with uncertainty, we are going to make use of the Wasserstein barycenters. Such an approach will allow us to get the probability measure that minimizes the 2-Wasserstein distances to interventional distributions of all possible subgraphs, where causal effect is identifiable, of a given causal graph. The subgraphs are going to be weighted according to three strategies in order to find the most reliable one.

Objective

The objective of this paper is to address the problem of uncertainty of relationships in causal models, by developing and evaluating a pipeline of algorithms for the Wasserstein barycenter approximation of causal effect $P(Z|do(X = x))$ on probabilistic causal graphs; to test it on different models, such as linear, binary, and non-linear ones, used for data generation process; to run a number of experiments on the proposed pipeline, changing certain parameters and subgraph-weighting strategy.

Tasks

To achieve this objective, the following tasks were identified:

- conduct literature research and explore causal inference, optimal transport problem as well as Wasserstein barycenters;
- design and implement an algorithm for splitting uncertain causal graphs into their surrogates;
- decide on the identifiability of causal effect in each of the surrogates;
- select and apply an appropriate estimation strategy (backdoor, frontdoor, instrumental variables) on the identifiable subset of surrogates;
- compute Wasserstein barycenter from the individual interventional distributions of subgraph estimates;
- build four DGP (data generation process) models: linear, binary (XOR), and two non-linear ones (polynomial, tanh); combined with weighting strategies, yielding six configurations $\mathcal{M}_1$-$\mathcal{M}_6$;
- compare three strategies for assigning weights to subgraph distributions before computing Wasserstein barycenter: random, probability of a subgraph, inverse of entropy of a subgraph;
- run a number of experiments on each model, modifying the parameter settings: sample size, intervention value, Bernoulli parameters for edge generation;
- assess the accuracy of the pipeline by computing the 2-Wasserstein distance between the target distribution and the barycenter;

Contributions

In this paper, the following contributions were made:

- Implementation of the algorithm for splitting the initial probabilistic graphs;

- Aggregation of algorithms for identifying, estimating causal effects, as well as computing the Wasserstein barycenters in semi-Markovian probabilistic graphs;
- Testing the pipeline of algorithms on different models, such as linear, binary, and non-linear;
- Comparing three subgraph-weighting strategies;
- Testing the influence of involved parameters.

Thesis Structure

The remainder of this thesis is structured as follows:

Preliminaries chapter presents the theoretical background.

Methods chapter describes the algorithms used for the pipeline.

Results chapter visualizes the outcome of experiments on various combinations of parameters used in the pipeline.

Discussion chapter elaborates and explains the outcomes from the previous chapter.

Conclusion chapter summarizes the work.

Preliminaries

Causal Inference

Structural Equation Models (SEMs)

Each variable in the network can be written as a structural equation $x_i = f_i(pa_i, u_i)$, consisting of its parents and the error term [1]. A set of such autonomous equations, where changes in one equation do not imply changes in others, is called a *structural equation model*.

Markovian Model

Since the time order of cause and its effect is essential, such relationships can be visualized by causal graphs, which capture the mentioned condition by providing an ancestral set for each of the variables in a model, i.e., the set of events preceding the specific one.

A directed acyclic graph (DAG), where an arrow from node X to node Y includes X into Y's ancestral set, i.e., sets X as a Markovian parent of Y, is called *Bayesian network* [1] . Having a Bayesian network of probability distribution P allows the use of decomposition[1]:

$$P(x_1, \ldots, x_n) = \prod_i P(x_i \mid pa_i) \tag{2.1}$$

A Bayesian network, where directed edges represent causal relationships between corresponding variables [5], is called *causal*. A causal Bayesian network allows the truncated factorization of $P_x(v)$ as a result of intervention [1]:

$$P_x(v) = \prod_{\{i|V_i \notin X\}} P(v_i \mid pa_i) \tag{2.2}$$

One of the properties of causal Bayesian network, defined in [1], namely $P_{pa_i,s}(v_i) = P_{pa_i}(v_i)$, ensures the autonomy of each $P(v_i \mid pa_i)$.

Causal Bayesian networks are also called a *Markovian models* [5]. The errors in such models are jointly independent. [1]

Causal models, in comparison to probability models, apart from showing the probability of events and their evolution in response to new observations, also show how these probabilities change due to external interventions [1].

Semi-Markovian Model

A causal acyclic graph $G(M)$ built from a causal model M - a description of the relationship between a variable and its parents, built upon a causal structure, based on the available observational data [1] - is called a *semi-Markovian model*. Both Markovian and semi-Markovian models are types of *structural causal models*, which, according to Pearl [1], are defined as a pair $M = \langle D, \Theta_D \rangle$, consisting of a causal structure D and a set of parameters Θ_D compatible with D.

While Markovian models consist of observed variables only, semi-Markovian models have unobserved common causes (confounding). Thus (2.2) no longer holds and a decomposition of interventional distribution into observational one becomes an identification problem, discussed further in the work.

Semi-Markovian models are going to be used in this work to represent causal graphs.

Interventional Distribution (do-operator)

An observational distribution is a passive analysis of a system, where a certain value of a variable updates beliefs about both descendants and predecessors. An *inter-*

ventional distribution shows an external intervention on the system, by forcing a certain value of a variable and hereupon deleting all connection with predecessors of the intervened variable. Since observational data is usually more accessible than interventional data due to ethical, economic, and feasibility constraints, the question that arises is whether it is possible to compute interventions using existing observations only. The answer to it is determined by output of the identifiability algorithm.

The difference between an observation $P(z \mid x)$ and an action $P(Z \mid do(X = x))$ from a graphical point of view lies in the way they are obtained: while the first one results from a Bayesian conditioning, the second one conditions after trimming all incoming edges to the intervened value [1].

Having a structural equation model (SEM) allows interventions to be computed by applying modifications to certain equations. An intervention $do(X = x)$ substitutes the equation for X with the value of intervention and modifies equations, containing it, accordingly.

Causal Effect

According to [1] definition, *causal effect*, denoted as $P(z \mid do(x))$, is a function from X to the space of probability distribution of Z, where X and Z are two disjoint sets of variables.

Identifiability of Causal Effects

As stated in the previous section, *identifiability* answers the question whether the interventional distribution $P(z \mid do(X = x))$ can be decomposed into an expression of observational distributions only, given a respective causal graph. Interventions can be computed from observational data only, having all variables of the identification set

measured, i.e., observed. The question of identifiability arises from the state where an identification set for a specific causal query has unmeasured, i.e., latent variables.

According to [1] *identifiability* describes the unambiguous interpretation of a computed causal quantity of interest $Q(M)$ in a particular causal model class $\boldsymbol{M}$: for any two models in the class that agree on $P(v)$, the quantity must take the same value [1]. Such classes fix the causal graph and assume positive distributions of observed variables, but may vary in the nature of parent-child relationships. It also ensures that the causal quantity is inferred from observational data and a causal diagram only.

Causal effect $P(z \mid do(X = x))$, as an example of such causal quantity, is identifiable from a graph G if it can be computed uniquely from any positive probability of the observed variables [1]. According to [9], if a particular causal query is identifiable, it means it can be expressed as a function of the observational distribution and thus estimated from observational data.

Do-Calculus

A set of three rules that are used to get rid of the *do*-operator in the initial causal query, given a causal graph, and switch to an expression, consisting purely of observational distributions, is called *do-calculus*. [12]

- Rule 1: $P_{\mathbf{x}}(\mathbf{y}|\mathbf{z},\mathbf{w}) = P_{\mathbf{x}}(\mathbf{y}|\mathbf{w})$ if $(\mathbf{Y} \perp\!\!\!\perp \mathbf{Z}|\mathbf{X},\mathbf{W})_{G_{\overline{\mathbf{x}}}}$
- Rule 2: $P_{\mathbf{x},\mathbf{z}}(\mathbf{y}|\mathbf{w}) = P_{\mathbf{x}}(\mathbf{y}|\mathbf{z},\mathbf{w})$ if $(\mathbf{Y} \perp\!\!\!\perp \mathbf{Z}|\mathbf{X},\mathbf{W})_{G_{\overline{\mathbf{x}},\underline{\mathbf{z}}}}$
- Rule 3: $P_{\mathbf{x},\mathbf{z}}(\mathbf{y}|\mathbf{w}) = P_{\mathbf{x}}(\mathbf{y}|\mathbf{w})$ if $(\mathbf{Y} \perp\!\!\!\perp \mathbf{Z}|\mathbf{X},\mathbf{W})_{G_{\overline{x},\overline{z(\mathbf{w})}}}$

According to [1], a causal effect is identifiable, if a finite sequence of the aforementioned rules allows the reduction to a *do*-free probability expression, consisting of observed quantities.

Confounding Variables

Confounding is the reason why the conditional distribution $P(Z \mid X)$ can not show the pure causal effect of X on Z, when there is a common cause behind both of the variables. Not taking confounding variables into consideration may result in reversed conclusions about the influence inside a particular relationship, as stated in Simpson's paradox [1].

Confounding - an unobserved common cause - is represented via a bidirected edge between two other nodes in causal diagrams. The following criteria describe certain structures in causal graphs, that enable elimination of *do*-operator in probability expressions and thus declare identifiability.

Back-door criterion.

Definition 2.1.1 [1] A set Z satisfies the back-door criterion relative to (X, Y) if

(i) no node in Z is a descendant of X, and

(ii) Z blocks every path between X and Y that contains an arrow into X.

$$P(y|do(x)) = \sum_{z} P(y|x, z)P(z). \tag{2.3}$$

Back-door is only applied when all variables in the adjustment set are observed, i.e., no latent confounding between X and Y.

Front-door criterion.

If the back-door criterion is not satisfied, i.e., the confounding is unobserved, a model may satisfy front-door criterion.

Definition 2.1.2 [1] A set Z satisfies the front-door criterion relative to (X, Y) if

(i) Z intercepts all directed paths from X to Y;

(ii) there is no unblocked back-door path from X to Z; and

(iii) all back-door paths from Z to Y are blocked by X.
If in addition $P(x, z) > 0$:

$$P(y \mid do(x)) = \sum_{z} P(z \mid x) \sum_{x'} P(y \mid x', z)\, P(x'), \tag{2.4}$$

The mediator must not be confounded with the intervened variable.

Back-door and front-door criteria are the consequences of the *do*-calculus rules mentioned in the previous section. In other words, patterns in graphs, whose *do*-calculus transformation is already defined.

Instrumental variables (IV).

If neither of the above criteria were met, IV tries an external variable.

Definition 2.1.3 [1] A variable Z is an instrument relative to the total effect of X on Y if Z only influences Y through X and Z is not independent of X.

Hedges

In previous section we defined graphical structures that declare identifiability. In this section, we take a look at a structure that results in unidentifiability.

Definition 2.1.4 (C-component [12]) Let G be a semi-Markovian graph such that a subset of its bidirected arcs forms a spanning tree over all vertices in G. Then G is a C-component (confounded component).

It means that all nodes in G are mutually reachable via bidirected edges. Any semi-Markovian graph is either a single C-component or can be partitioned into a set of such.

Definition 2.1.5 (C-forest [12]) Let G be a semi-Markovian graph, where Y is the root set. Then G is a Y-rooted C-forest (confounded forest) if G is a C-component, and all observable nodes have at most one child.

The root set is a set of childless nodes, with $An(Y)_G = G$, where $An(Y)_G$ is the ancestry set of Y.

Definition 2.1.6 (Hedge [12]) Let X, Y be disjoint sets of variables in G. Let F, F' be $\mathbf{R}$-rooted C-forests such that $F \cap \mathbf{X} \neq \emptyset$, $F' \cap \mathbf{X} = \emptyset$, $F' \subseteq F$, and $\mathbf{R} \subset An(\mathbf{Y})_{G_{\bar{\mathbf{x}}}}$. Then F and F' form a hedge for $P_{\mathbf{x}}(\mathbf{y})$ in G.

Hedge - a special case, formed by two C-forests - is what causes the unidentifiability of causal effect in a graph that contains it with regard to a certain query.

Optimal Transport and Barycenters

Optimal Transport Problem

The question of distance between probability measures arises in multiple domains from Machine learning and statistics to physics. Instead of calculating the distance in particular, one can size the effort of reshaping one measure into the other, namely to find an optimal transport. The initial version of this problem was formulated by Monge (6.0.1 [10]):

$$\inf \left\{ \int_X c(x, \boldsymbol{t}(x))\, d\mu(x) : \boldsymbol{t}_{\#}\mu = \nu \right\}, \tag{2.5}$$

where X represents a source metric space, μ is the measure on the source space, which tells how much mass is in the selected region in source space, ν is the measure on the target space, c is a cost function and $\boldsymbol{t}$ is a transport map. This formulation was later enhanced by Kantorovich to overcome cases, where no transport map exists due to a limitation that no mass can be split. In this work, we focus on the Kantorovich optimal transportation problem, which can always find a feasible solution:

$$\min \left\{ \int_{X \times Y} c(x, y)\, d\boldsymbol{\gamma}(x, y) : \boldsymbol{\gamma} \in \Gamma(\mu, \nu) \right\}, \tag{2.6}$$

here the transport map is substituted by an optimal transport plan γ.

Wasserstein Distance

According to [10] , the p-th Wasserstein distance between two probability measures with regard to p is:

$$W_p^p(\mu^1, \mu^2) := \min \left\{ \int_{X^2} d(x_1, x_2)^p\, d\boldsymbol{\mu}(x_1, x_2) : \boldsymbol{\mu} \in \Gamma(\mu^1, \mu^2) \right\}. \tag{2.7}$$

In order to get the pure distance, we take the p-th root of (2.7) and receive [6]:

$$W_p(\mu,\nu) \stackrel{\text{def}}{=} \left(\inf_{\pi\in\Pi(\mu,\nu)} \int_{\Omega^2} D(x,y)^p \, d\pi(x,y)\right)^{1/p}. \tag{2.8}$$

Wasserstein Space

According to [11] :

$$P_p(\mathcal{X}) := \left\{\mu \in P(\mathcal{X}); \qquad \int_{\mathcal{X}} d(x_0,x)^p \, \mu(dx) < +\infty\right\}, \tag{2.9}$$

the Wasserstein space is defined as "space of probability measures which have a finite moment of order p", having a finite Wasserstein distance. In this work, $p = 2$, which corresponds to a finite second moment - can be viewed as variance - meaning the spread of probability distributions in the space is finite.

Wasserstein Barycenter

In this work, we are aiming to calculate an average over the probability distributions of estimates of individual subgraphs as an approximation of causal effect in the whole causal graph. According to [7] the barycenter of the measures ν_i's with weights λ_i's is the solution to the following minimization problem:

$$\inf_{\nu} \sum_{i=1}^{p} \lambda_i W_2^2(\nu_i,\nu), \tag{2.10}$$

where $\lambda_i > 0$ and $\sum_{i=1}^{p} \lambda_i = 1$.

Definition 2.2.1 ([7]) the Wasserstein barycenter of a set of probability measures $(\nu_1,\ldots,\nu_p)$ with weights $(\lambda_1,\ldots,\lambda_p)$, summing to 1, is the solution of the problem above.

Methods

This chapter walks through the whole pipeline of algorithms, used to estimate causal effects based on causal graphs and observational data, as well as explains how the aforementioned concepts are implemented.

DGP (Data Generation Process)

The data generation process consists of two subsequent steps: graph generation and dataset generation. After fixing the number of observed nodes, one needs to fix two independent probabilities, q and p, which are used as Bernoulli parameters for the random generation of directed and bidirected edges, respectively. The structure of the resulting graph is stored in a form of an adjacency matrix. In order to keep the resulting graph aligned with the DAG properties, it was chosen to make use of strictly lower triangular matrices. Apart from the adjacency matrix, the algorithm stores edge probabilities as well as produces a matrix of coefficients (in case of linear and non-linear models), used for future dataset generation.

The dataset is generated taking the number of samples N, an adjacency matrix, and a corresponding coefficient matrix (for non-binary models). The value of a node in each sample is defined as a combination of its parents and a respective independent noise. The exact formula is defined based on the model: linear, binary, or non-linear.

Formula for $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$:

$$X_i = \sum_{X_j \in Pa(X_i)} \beta_{ji} X_j + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1)$$

Formula for $\mathcal{M}_4$:

$$X_i = \left(\bigoplus_{X_j \in Pa(X_i)} X_j \right) \oplus \varepsilon_i, \quad \varepsilon_i \sim \text{Bernoulli}(0.1)$$

Formula for $\mathcal{M}_5$:

$$X_i = \sum_{X_j \in Pa(X_i)} \left(\alpha_{ji} X_j^2 + \beta_{ji} X_j \right) + \varepsilon_i$$

Formula for $\mathcal{M}_6$:

$$X_i = \tanh\left(\sum_{X_j \in Pa(X_i)} \beta_{ji} X_j \right) + \varepsilon_i$$

Formula for noise in $\mathcal{M}_5$ and $\mathcal{M}_6$:

$$\varepsilon_i \sim \tfrac{1}{2}\mathcal{N}(-2,\, 0.5^2) + \tfrac{1}{2}\mathcal{N}(2,\, 0.5^2)$$

The algorithm returns a dataset consisting of a particular number of rows, which corresponds to the number of nodes in a corresponding graph, and N number of columns.

Splitting Causal Graph into Subgraphs

This research aims to estimate causal effect in each possible scenario of a probabilistic graph structure in order to average them afterwards using Wasserstein barycenters. Thus the previously generated graph has to be split into all possible subgraphs. The number of such subgraphs corresponds to 2^m, where m is the number of uncertain edges, i.e. probabilities of which lie between 0 and 1 exclusively. Each subgraph corresponds to one of the combinations of m-fold Cartesian product $\{True, False\}^m$.

Estimating Causal Effect

In order to estimate causal effect of an intervention for a specific subgraph, the algorithm relies on the *DoWhy* Python library, in particular on its *CausalModel*.

After taking the subdataset, which consists of observed nodes only, the subgraph structure as well as treatment and outcome variables, the model decides on the identifiability of the query. If causal effect is identifiable in a particular subgraph, it proceeds by using one of the three strategies: back-door, front-door, or instrumental variables. After calculating an estimate, it is stored alongside the weight of a subgraph. The weights are calculated in one of three ways: randomly generated, calculated as a product of probabilities of all edges in a subgraph that is being processed, or as an inverse to the conditional entropy of a subgraph, i.e., how predictable the target value is, based on its observed parents in a particular subgraph. All weights sum up to 1.

Finding Wasserstein Barycenters and Comparing with the True Distribution

The Wasserstein barycenter is a distribution that represents the center of previously calculated probability distributions of estimates of each subgraph. Thus, it aims to be close to the true target distribution, calculated using the whole dataset, including unobserved nodes. The algorithm uses the Python library *POT* (Python Optimal Transport), which focuses on solving optimal transport problems.

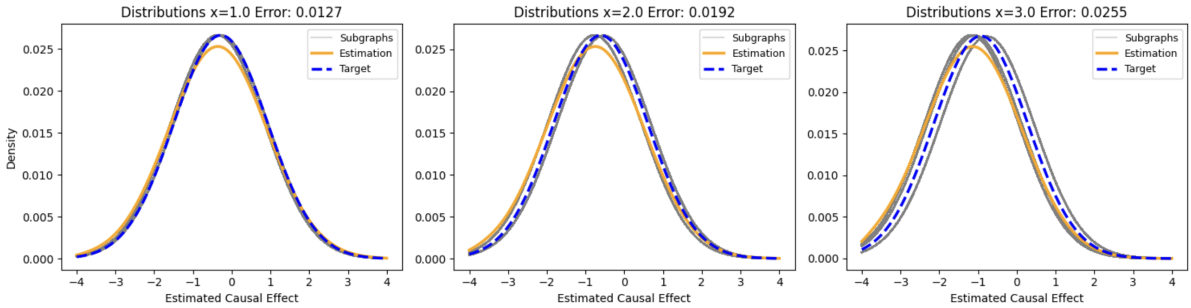
All distributions are spread over the grid y of n points, which represent the range of possible values of estimates. The algorithm requires a cost matrix $M \in \mathbb{R}^{n \times n}$, where M_{ij} stores the cost of moving from point y_i to point y_j. The cost is defined as *squared Euclidean* distance between two points on the grid. M is normalized by its maximum entry, so that costs are expressed relative to the grid width. In order to compare the true target distribution with the resultant barycenter distribution, 2-Wasserstein distances were calculated.

Results

Linear Model with Weights = Probability of Subgraph ($\mathcal{M}_1$)

Figure 1

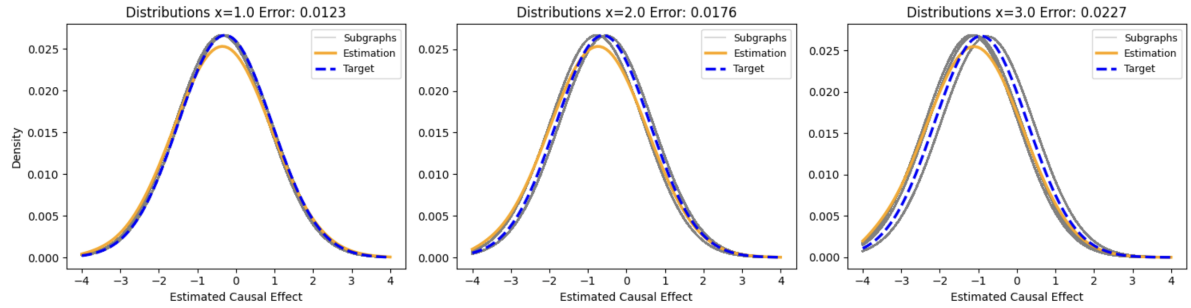
6 Observed nodes. Parameters $q = 0.7$, $p = 0.2$ and $N = 5000$ are fixed. X varies.



Linear Model with random Weights ($\mathcal{M}_2$)

Figure 2

6 Observed nodes. Parameters $q = 0.7$, $p = 0.2$ and $N = 5000$ are fixed. X varies.



Linear Model with Weights = Inverse of Entropy ($\mathcal{M}_3$)

Figure 3

6 Observed nodes. Parameters $q = 0.7$, $p = 0.3$ and $N = 10000$ are fixed. X varies.

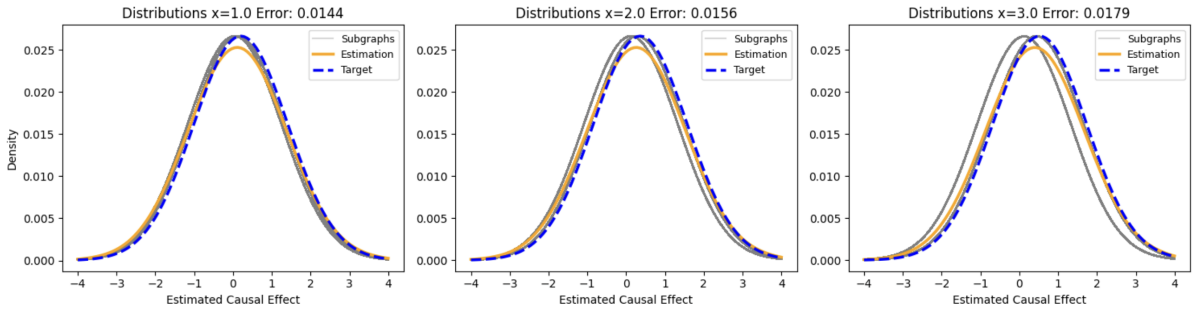


Figure 4

6 Observed nodes. Parameters $q = 0.7$, $p = 0.2$ are fixed. N varies.

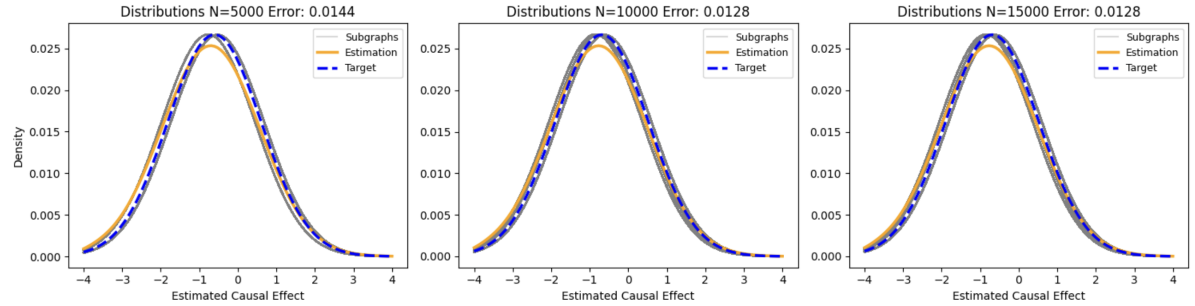


Figure 5

5 Observed nodes. Parameters $N = 7000$, $q = 0.9$, $X = 1.0$ are fixed. p varies.

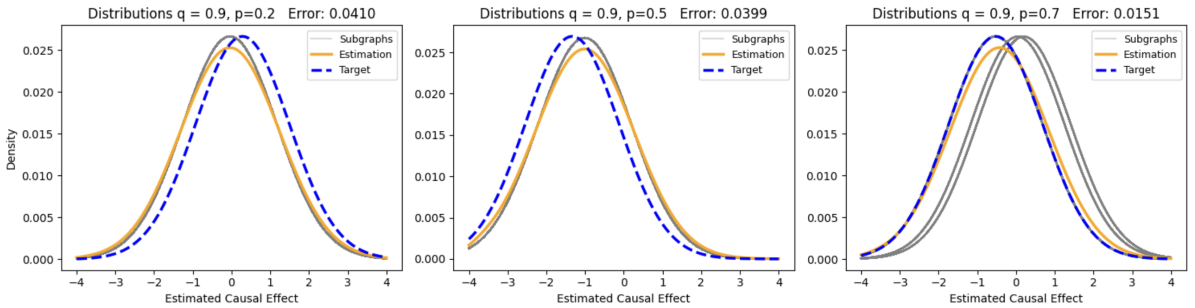
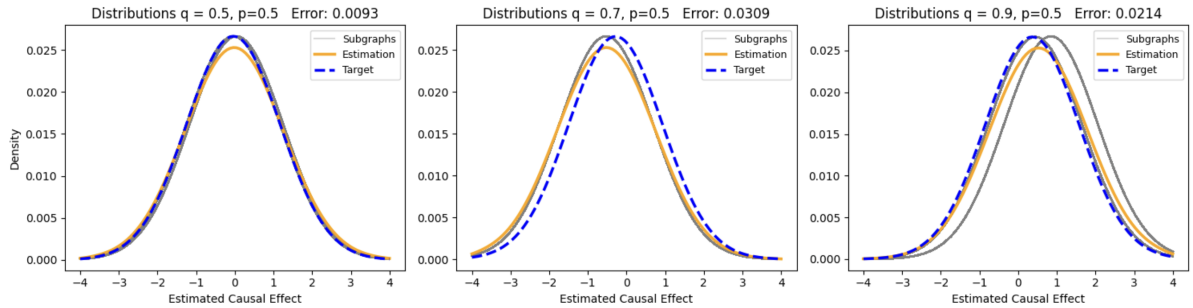


Figure 6

5 Observed nodes. Parameters $N = 7000$, $p = 0.5$, $X = 1.0$ are fixed. q varies.



Binary Model ($\mathcal{M}_4$)

Figure 7

7 Observed nodes. Parameters $q = 0.7$, $p = 0.2$ are fixed. N varies.

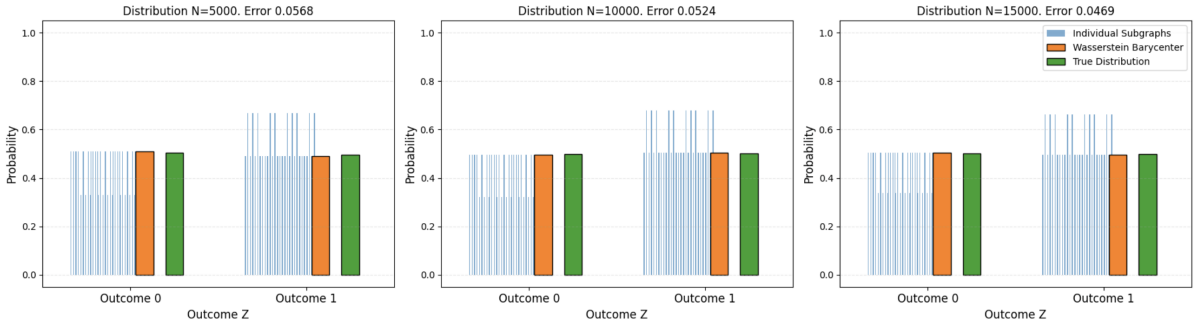


Figure 8

5 Observed nodes. Parameters $q = 0.9$, $N = 7000$ are fixed. p varies.

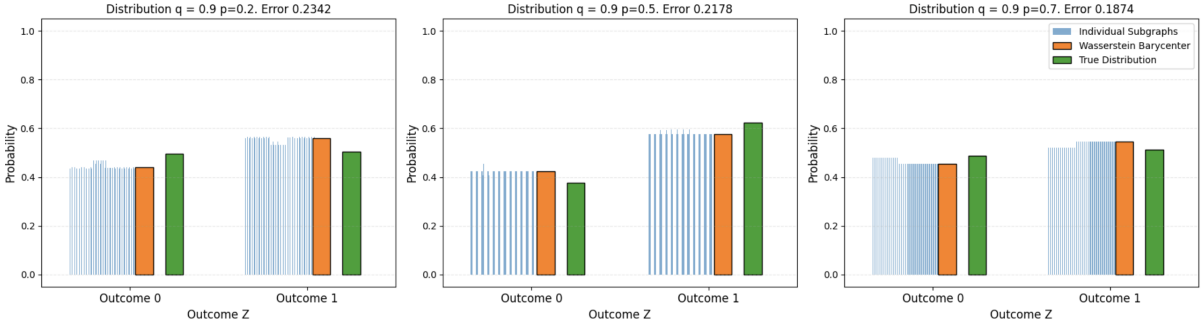
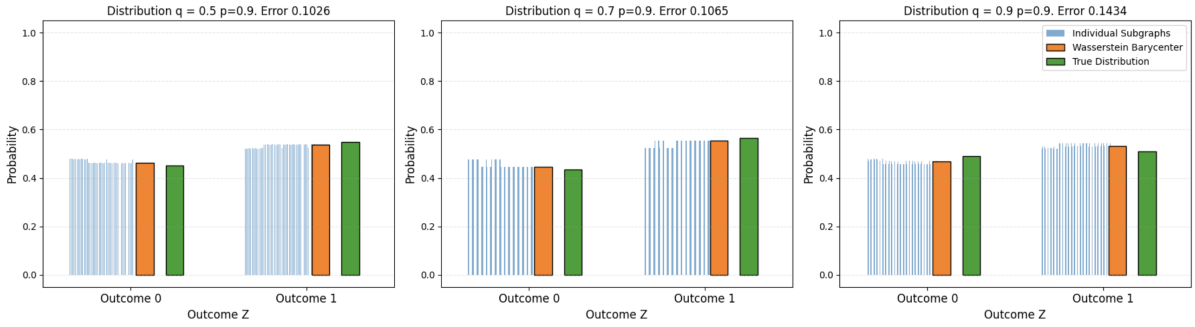


Figure 9
5 Observed nodes. Parameters $p = 0.9$, $N = 7000$ are fixed. q varies.



Non-Linear Model (polynomial) ($\mathcal{M}_5$)

Figure 10
5 Observed nodes. Parameters $q = 0.5$, $p = 0.2$ and $N = 10000$ are fixed. X varies.

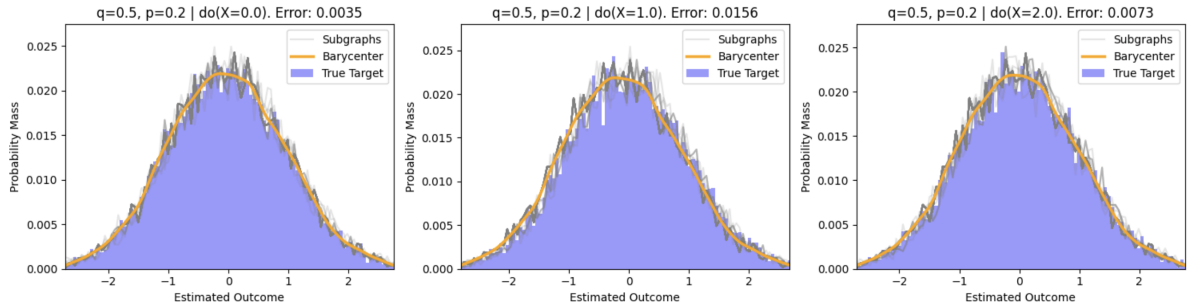


Figure 11
6 Observed nodes. Parameters $q = 0.7$, $p = 0.2$ and $X = 2.0$ are fixed. N varies.

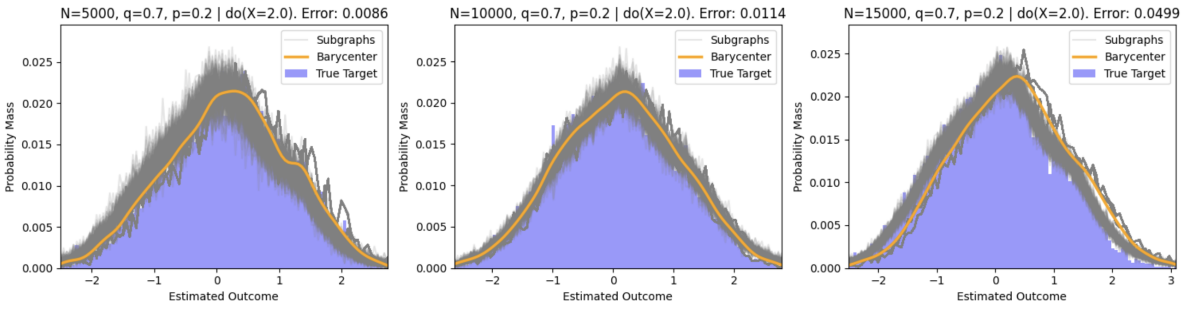


Figure 12
5 Observed nodes. Parameters $q = 0.9$, $X = 2.0$ and $N = 7000$ are fixed. p varies.

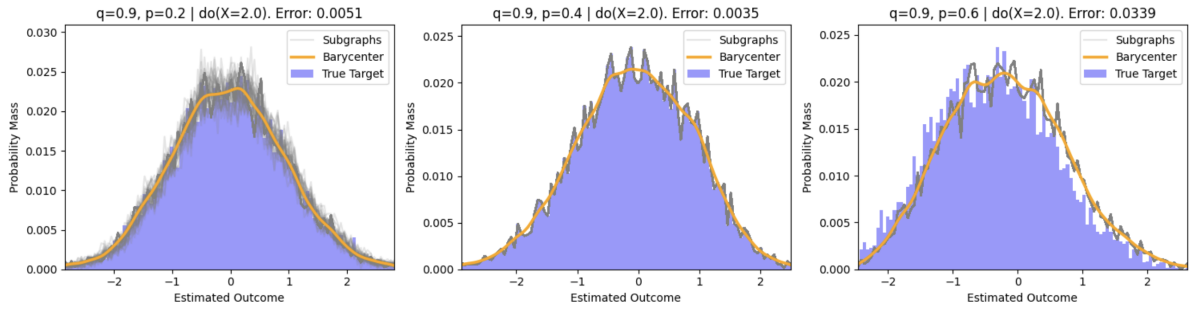
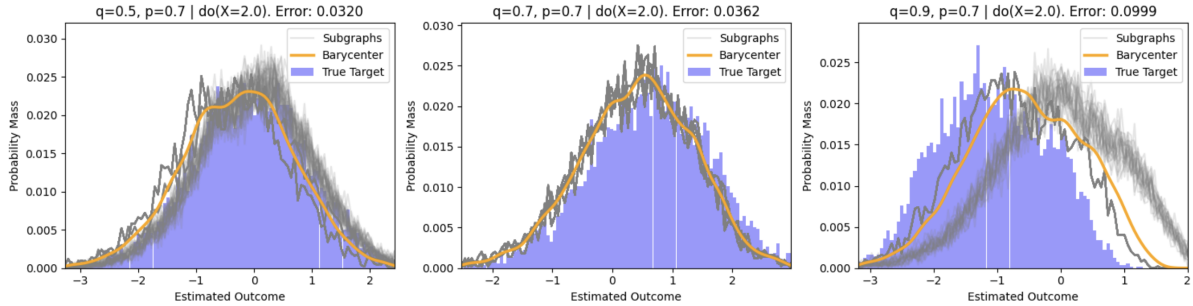


Figure 13
5 Observed nodes. Parameters $p = 0.7$, $X = 2.0$ and $N = 5000$ are fixed. q varies.



Non-Linear Model (tanh) ($\mathcal{M}_6$)

Figure 14
5 Observed nodes. Parameters $q = 0.5$, $p = 0.2$ and $N = 10000$ are fixed. X varies.

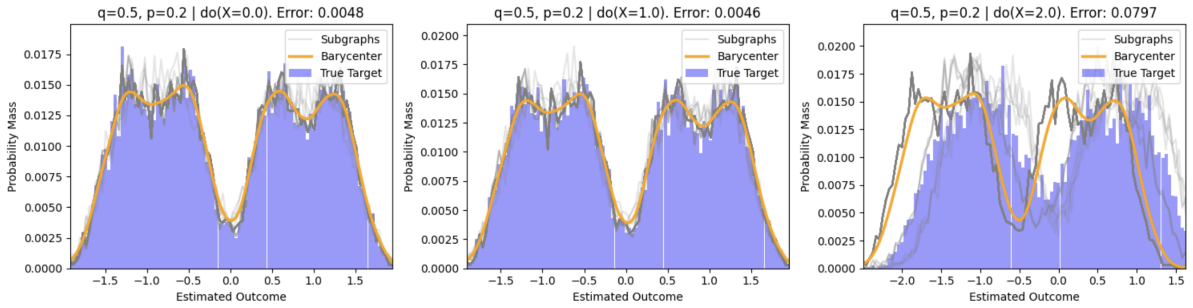


Figure 15

6 Observed nodes. Parameters q = 0.7, p =0.2 and X = 2.0. N varies.

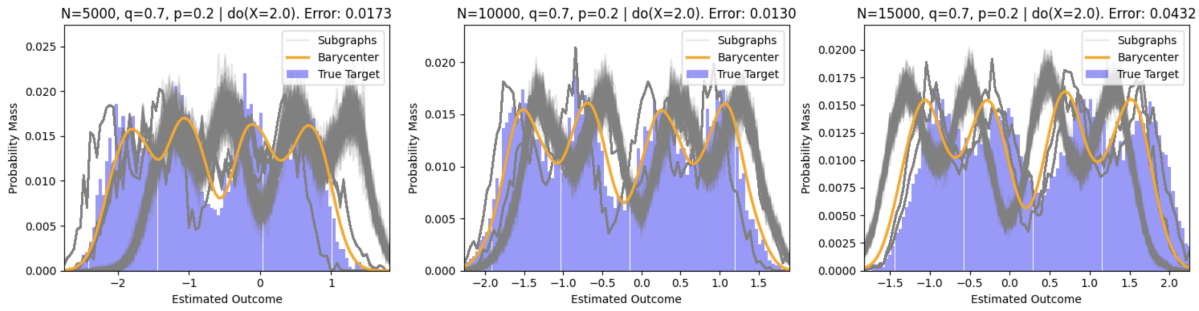


Figure 16

5 Observed nodes. Parameters q = 0.9, X = 2.0 and N = 7000 are fixed. p varies.

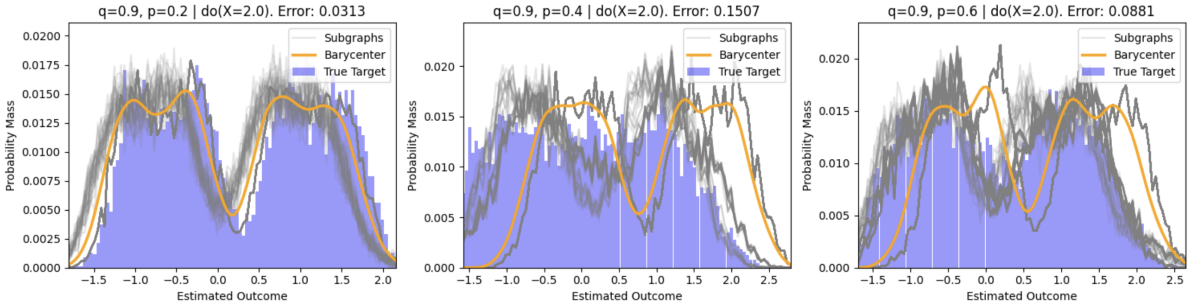
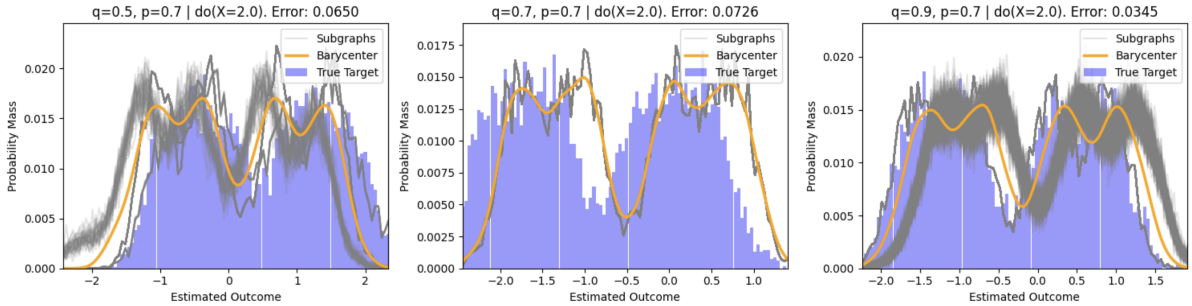


Figure 17

5 Observed nodes. Parameters p = 0.7, X = 2.0 and N = 7000 are fixed. q varies.



Representative single panels:

Figure 18
6 Observed nodes. Parameters $q = 0.7$, $p = 0.2$, $X = 3.0$ and $N = 5000$. $\mathcal{M}_1$

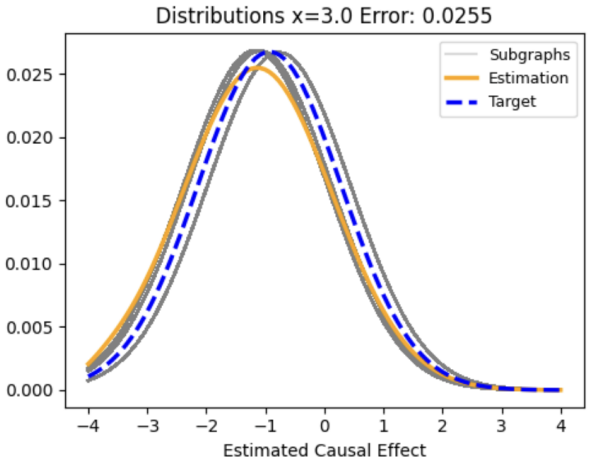


Figure 19
6 Observed nodes. Parameters $q = 0.7$, $p = 0.2$, $X = 3.0$ and $N = 5000$. $\mathcal{M}_2$

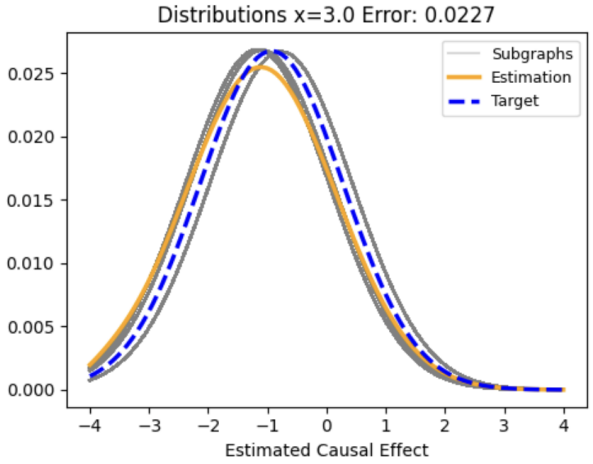


Figure 20

6 Observed nodes. Parameters $q = 0.7$, $p = 0.2$, $X = 3.0$ and $N = 5000$. $\mathcal{M}_3$

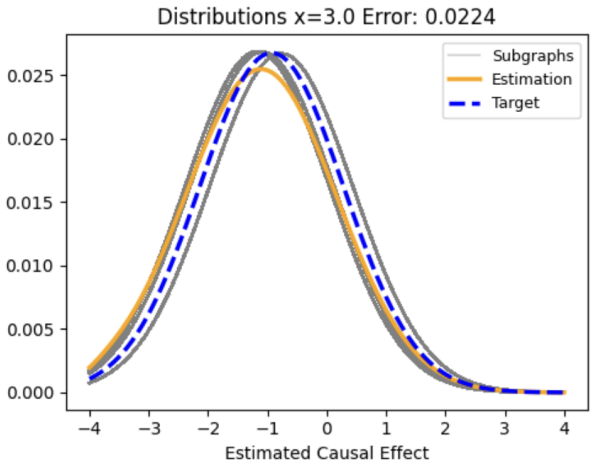


Figure 21

7 Observed nodes. Parameters $q = 0.7$, $p = 0.2$ and $N = 15000$. $\mathcal{M}_4$

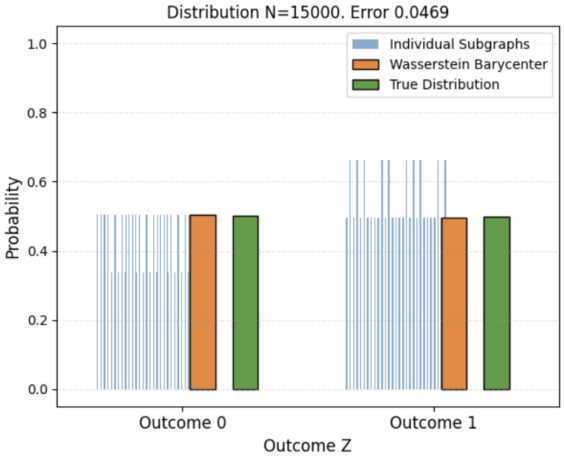


Figure 22
5 Observed nodes. Parameters $q = 0.9$, $p = 0.2$ and $N = 7000$. $\mathcal{M}_5$

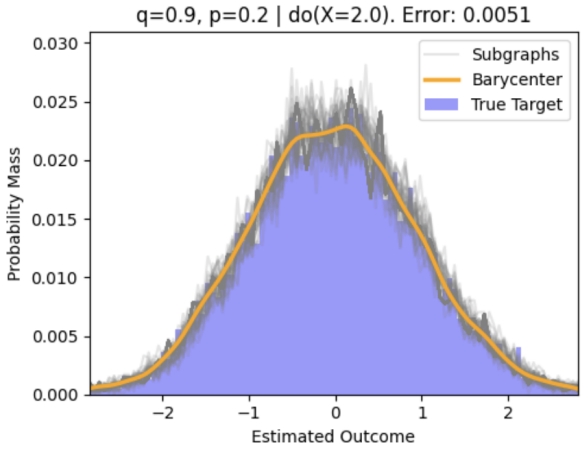


Figure 23
5 Observed nodes. Parameters $q = 0.9$, $p = 0.2$ and $N = 7000$. $\mathcal{M}_6$

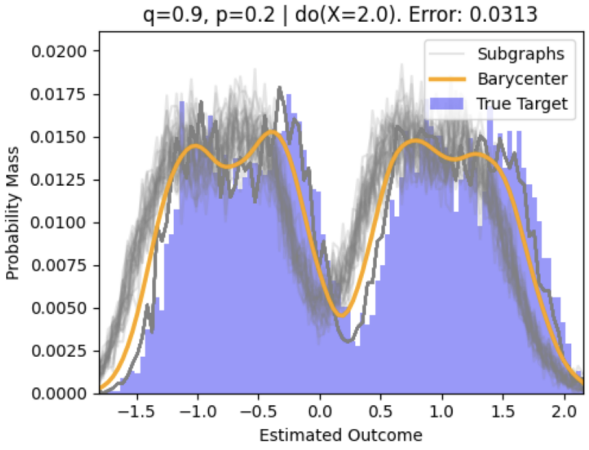


Table 1
2-Wasserstein distances $W_2(\mathbb{P}_x(Z), \cdot)$ between the true interventional distribution $\mathbb{P}_x(Z)$ and three comparison distributions, across the six models $\mathcal{M}_1$-$\mathcal{M}_6$. $\mathbb{P}_1$ and $\mathbb{P}_2$ are the two individual subgraph surrogates; $\mathbb{P}_b$ is the Wasserstein barycenter.

$W_2(\mathbb{P}_x(Z), \cdot)$	$\mathbb{P}_1$	$\mathbb{P}_2$	$\mathbb{P}_b$
$\mathcal{M}_1$	0.020	0.029	**0.026**
$\mathcal{M}_2$	0.020	0.029	**0.023**
$\mathcal{M}_3$	0.020	0.030	**0.022**
$\mathcal{M}_4$	0.405	0.047	**0.047**
$\mathcal{M}_5$	0.006	0.007	**0.005**
$\mathcal{M}_6$	0.061	0.016	**0.031**

The table above represents *2-Wasserstein* distances between the true interventional distribution $P(Z \mid do(X = x))$, computed on the whole dataset, including unobserved variables, and the three comparison distributions of the representative single panels Figures 18–23. Surrogate distributions $\mathbb{P}_1$ and $\mathbb{P}_2$ are sampled at fixed positions in subgraph ordering of each model. $\mathbb{P}_b$ is the Wasserstein barycenter approximation of the estimate, calculated using all identifiable subgraphs. Since W_2 is the distance to the target, representing the error in approximation, smaller values represent better results.

Discussion

The main task of this thesis was to build a pipeline of algorithms for approximating causal effects, i.e., the interventional distribution $P(Z \mid do(X))$ of probabilistic causal graphs using Wasserstein barycenters. In order to validate the aforementioned method in various scenarios, six models, which vary either in subgraph-weighting or the DGP strategies, were built. The results, visualized in the previous chapter, are going to be analyzed here. The errors, which are used to assess the quality of approximation, are computed as the 2-Wasserstein distance between the resulting barycenter of identifiable subgraphs and the true interventional distribution, where all variables are observed.

Weights of Subgraphs

The first three models $\mathcal{M}_1$, $\mathcal{M}_2$, $\mathcal{M}_3$, presented on Figures 18–20 respectively, vary in the subgraph-weighting strategy only. Though the difference between error terms is relatively small, the possible reasons are worth discussing.

$\mathcal{M}_1$ uses the probability of a subgraph, calculated as a product of probabilities of present edges, and results in a 0.0255 error value. $\mathcal{M}_2$ assigns random values between 0 and 1, sampled from a uniform distribution, and results in a 0.0227 error value. Even though in both strategies weights are being normalized before calculating barycenters, the resulting array of weights differs. While $\mathcal{M}_2$ has relatively evenly spread values of weights after normalization, $\mathcal{M}_1$ might have a skewed distribution, dominated by sparse subgraphs with a small number of edges. The reason for that: a smaller number of values between 0 and 1 are being multiplied, resulting in a bigger overall product. The difference between values before normalization can differ by several orders of magnitude, which is preserved after normalization. Since not all sparse subgraphs produce reliable estimates, such a strategy produces a slightly higher error.

$\mathcal{M}_3$, which has the lowest error, namely 0.0224, uses the most reliable approach, especially for real-world scenarios, where graphs are not randomly generated, and probabilities of causal relationships are not known - the inverse of the conditional entropy of a subgraph. Such a strategy ensures that the weight of a subgraph corresponds to how well the target variable can be predicted by its parents, assigning larger weights to more informative subgraphs. In subsequent sections, the subgraph-weighting strategy is fixed at the inverse of conditional entropy.

Number of Samples

N - the number of samples in a dataset - varies among 5000, 10000 and 15000. The linear model $\mathcal{M}_3$ as well as the binary one $\mathcal{M}_4$ show a gradual decrease in error with the increase of N on Figures 4 and 7, respectively. Such trend may indicate that the model can derive a more precise estimate, having more data. The estimation process, on the contrary to the identification process, which relies purely on the graph structure, uses the dataset to evaluate the expression given by the identification algorithm. Non-linear model $\mathcal{M}_5$ shows a rather opposite tendency, while $\mathcal{M}_6$ delivers fluctuating results.

Value of Interventions

In linear models $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_3$, the lower the intervention value, the lower the error term. The reason for such behaviour lies in the linear structure of the estimate calculation, which is: $ATE * x$, where ATE is the average treatment effect (since the mean of initial target distribution is zero, the intercept term can be neglected here). The gap between the true expected value of the target and the approximation is thus $(ATE_t - ATE_e) * x$. The X value scales the gap between the estimated ATE and the true ATE.

For the binary model $\mathcal{M}_4$, such an experiment was not relevant, since the only meaningful value for intervention is $X = 1$; thus, the estimate is $P(Z = 1 \mid do(X = 1))$.

For the non-linear models, no such trend, as with linear ones, was captured, since the intervention effect is not proportional to the value of x in a non-linear, non-monotone function.

Bernoulli Parameters

Parameter q is responsible for whether or not a directed edge from one node to another exists. Parameter p is responsible for whether or not a bidirected edge between two nodes, i.e., an unobserved node pointing to two nodes, exists.

The linear model in Figure 5 shows that with a fixed $q = 0.9$, which already means dealing with a dense graph, the growing number of bidirected edges and thus unobserved variables leads to a decrease in the 2-Wasserstein error between a barycenter and a true distribution. Fixing $p = 0.5$ to a moderate value and varying q between 0.5, 0.7 and 0.9, on the contrary, does not show any particular tendency.

Taking a look at the binary model on Figure 8, we can see the same pattern for varying p-value: the error gradually decreases. Varying q, however, with a high number of unobserved nodes, namely $p = 0.9$, leads to an increase in the error term.

The non-linear model $\mathcal{M}_5$ demonstrates the same pattern in Figure 13 for q as in the binary case: the growing density of a graph in terms of directed edges increases the 2-Wasserstein error. The error fluctuates as p varies in a dense graph with fixed $q = 0.9$, as it is shown in Figure 12.

The non-linear model $\mathcal{M}_6$ in Figure 16 shows the fluctuations of error in both varying p and q. In both cases, the error term first goes up and then goes down.

Conclusion

Causal inference, in particular the estimation of causal effects, finds great use in various spheres, where it is important to link the cause with the effect: from medicine and agriculture to finance and machine learning. In complex real-world scenarios, multiple confounding factors must be accounted for in order to get an accurate estimation. Moreover, the uncertainty of (causal) relationships in an investigated causal model adds complexity to the problem. This work focused on a way of combining the identification algorithm as the causal identification step of causal inference with probabilistic causal graphs, representing the uncertainty of real-world complex scenarios.

The core idea is to split the causal graph into all possible subgraphs, based on the probabilities of edges, where each subgraph represents a possible surrogate graph; identify and estimate the causal effect $P(z \mid do(X = x))$ of each identifiable subgraph, based on the observational data, and compute the weighted center of probability measures - the Wasserstein barycenter - which represents the estimate of causal effect of the initial probabilistic graph.

After exploring the relevant and foundational literature on the topic, a pipeline of algorithms was implemented and aggregated. This includes: data and graph generation, splitting a graph into subgraphs, identification and estimation of causal effect in each subgraph, computing the Wasserstein barycenter, and calculating the 2-Wasserstein distance from the resulting barycenter to the true distribution. The pipeline was tested on several models: linear, binary, and two non-linear ones.

The Wasserstein barycenter appeared to be a reliable approximation of causal effect in an uncertain probabilistic graph in all of the observed models, resulting in a 2-Wasserstein error of less than 5% of the width of the grid of estimates (1) in

representative single panels Figures 18–23. Across all models, the barycenter captured the true interventional distribution even when the individual subgraph estimates were widely scattered. The effect is clearest for the binary model $\mathcal{M}_4$: a single subgraph surrogate has a 2-Wasserstein error 0.405, while the barycenter of the same subgraphs achieved 0.047 (1).

Multiple experiments on the involved parameters as well as the subgraph-weighting approaches have resulted in a number of observations in particular cases, namely how density of the graph, number of samples, and the intervention value influence the 2-Wasserstein distance.

Experiments were limited to generated datasets, containing between 5000 and 15000 samples, 5-7 observed nodes, and evaluated using one random seed. Averaging over several seeds and using larger graphs is a direction of future work. The implementations and findings of this work may further be tested on real data, not generated for research purposes.

Causal inference has enabled a deeper understanding of causal relationships between events in conditions where solely observational data is available. However, it is followed by a number of challenges. One such challenge, namely dealing with uncertainty of causal graph, was addressed and approached in this work, resulting in meaningful estimations and demonstrating the potential of the Wasserstein barycenter approximation of causal effects.

I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 03.08.2026 Anastasiia Kryvoruchko

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